

$$(3I_2)A = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 & -1 \\ 5 & -2 \end{bmatrix} = \begin{bmatrix} (3I_2)a_1 & (3I_2)a_2 \end{bmatrix} \text{ (By defn. of matrix multiplication)}$$

$$(3I_2)a_1 = \begin{bmatrix} 3 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 4 \\ 5 \end{bmatrix} = \begin{bmatrix} 3 \cdot 4 + 0 \cdot 5 \\ 0 \cdot 4 + 3 \cdot 5 \end{bmatrix} \text{ (Row-Vector Rule)}$$

$$= \begin{bmatrix} 12 \\ 15 \end{bmatrix} = (-1) \begin{bmatrix} 3 \\ 0 \end{bmatrix} + (-2) \begin{bmatrix} 0 \\ 3 \end{bmatrix} \text{ (By defn. of } Ax)$$

$$= \begin{bmatrix} -3 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -6 \end{bmatrix} \text{ (By defn. of scalar multiple)}$$

$$= \begin{bmatrix} -3 \\ -6 \end{bmatrix} \text{ (By defn. of vector sum)}$$

$$\therefore (3I_2)A = \begin{bmatrix} 12 & -3 \\ 15 & -6 \end{bmatrix}$$

In Exercises 5 and 6, compute the product AB in two ways: (a) by the definition, where Ab_1 and Ab_2 are computed separately, and (b) by the row-column rule for computing AB .

$$5. A = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix}, B = \begin{bmatrix} 3 & -2 \\ -2 & 1 \end{bmatrix}$$

$$a) AB = \begin{bmatrix} Ab_1 & Ab_2 \end{bmatrix} \text{ (By defn. of matrix multiplication)}$$

$$Ab_1 = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \end{bmatrix} = 3 \begin{bmatrix} -1 \\ 5 \\ 2 \end{bmatrix} + (-2) \begin{bmatrix} 2 \\ 4 \\ -3 \end{bmatrix} \text{ (By defn. of } Ax)$$

$$= \begin{bmatrix} -3 \\ 15 \\ 6 \end{bmatrix} + \begin{bmatrix} -4 \\ -8 \\ 6 \end{bmatrix} \text{ (By defn. of scalar multiple)}$$

$$= \begin{bmatrix} -7 \\ 7 \\ 12 \end{bmatrix} \text{ (By defn. of vector sum)}$$

~~$$Ab_2 = \begin{bmatrix} -1 & 2 \\ 5 & 4 \\ 2 & -3 \end{bmatrix} \begin{bmatrix} -2 \\ 1 \end{bmatrix} = \begin{bmatrix} (-1)(-2) + 2(1) \\ 5(-2) + 4(1) \\ 2(-2) + (-3)(1) \end{bmatrix} = \begin{bmatrix} 4 \\ -6 \\ -7 \end{bmatrix}$$~~